Summary in Graph

# Exam Summary (GO Classes CS/DA 2026 | Linear Algebra| Weekly Quiz 2)

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|---|---|---|---|
| Qs. Attempted: | 0 (0 + 0) | Correct Marks: | 0 (0 + 0) |
| Correct Attempts: | 0 (0 + 0) | Penalty Marks: | 0 (0 + 0) |
| Incorrect Attempts: | 0 (0 + 0) | Resultant Marks: | 0 (0 + 0) |

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|---|---|
| Total Questions: | 10 (10 + 0) |
| Total Marks: | 10 (10 + 0) |
| Exam Duration: | 30 Minutes |
| Time Taken: | 0 Minutes |

## Individual Question Statistics

Select Metric: Marks Obtained

Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10

* Average Stats based on first-attempt stats of Top 10% (141) out of 1412 users

EXAM RESPONSE EXAM STATS FEEDBACK

## Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Linear Algebra

Let $\mathbf{v}$ be a solution to $A\mathbf{x} = \mathbf{0}$ and $\mathbf{w}$ be a solution to $A\mathbf{x} = \mathbf{b}$, where $\mathbf{b} \neq \mathbf{0}$. Find all scalars $r$ and $s$ such that $r\mathbf{v} + s\mathbf{w}$ is a solution to $A\mathbf{x} = \mathbf{b}$.

A. $r = 1$, and $s = 1$ are the only possible values of scalars
B. $r = 0$ and $s$ can be anything
C. $r$ can be anything, and $s = 1$

D. $r$ and $s$ can both be any scalar

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider the following statements about linear systems and matrices. Determine which ones are TRUE.

A. If $v_1, v_2$, and $v_3$ are vectors in $\mathbf{R}^2$, then the vector equation $x_1v_1 + x_2v_2 + x_3v_3 = \binom{0}{0}$ must have infinitely many solutions.
B. Suppose $A$ is a $3 \times 2$ matrix and $b$ is a vector in $\mathbf{R}^3$ such that the equation $A\mathbf{x} = \mathbf{b}$ has exactly one solution. Then, the homogeneous system $A\mathbf{x} = 0$ must have only the trivial solution.
C. If $A$ is a $4 \times 5$ matrix and the solution set to $A\mathbf{x} = 0$ is a line, then the system $A\mathbf{x} = \mathbf{b}$ must be inconsistent for some $\mathbf{b} \in \mathbf{R}^4$.
D. A $2 \times 2$ matrix $A$ exists such that the solution set of $A\mathbf{x} = \binom{1}{2}$ is the point $\left\{\binom{-1}{1}\right\}$, and the system $A\mathbf{x} = \binom{2}{1}$ is inconsistent.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Suppose that the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4 \in \mathbb{R}^4$ span $\mathbb{R}^4$. Let $A$ be the $4 \times 4$ matrix having $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ as columns. If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^4$ are vectors such that $A\mathbf{x} = A\mathbf{y}$, then which of the following is true?

A. $\mathbf{x} = \mathbf{y}$
B. $\mathbf{x} - \mathbf{y} = \mathbf{v}_1$
C. $\mathbf{x} - \mathbf{y}$ is a nonzero solution to $A\mathbf{z} = \mathbf{0}$
D. $A\mathbf{x} = A\mathbf{y}$ implies $A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$, but $\mathbf{x} \neq \mathbf{y}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider the following statements about linear systems. Which of the following are TRUE?

A. If an $m \times n$ matrix $A$ has $m$ pivots after row reduction, then for each $\mathbf{b} \in \mathbb{R}^m$, the matrix equation $A\mathbf{x} = \mathbf{b}$ has at least one solution.
B. If a system of linear equations $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions, then the system of linear equations $A\mathbf{x} = \mathbf{c}$ for any other vector $\mathbf{c}$ has either no solution or infinitely many solutions.
C. For three vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$, if $\{\mathbf{u}, \mathbf{v}\}$ is linearly independent, and $\{\mathbf{v}, \mathbf{w}\}$ is linearly independent, then $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly independent.
D. If $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ are linearly dependent, then $\mathbf{u}$ is a linear combination of $\mathbf{v}$ and $\mathbf{w}$.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 06sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider the matrix $A$ with the following facts:

$$A\begin{bmatrix}-1\\1\end{bmatrix}=\begin{bmatrix}-4\\3\end{bmatrix}$$
$$A\begin{bmatrix}1\\-2\end{bmatrix}=\begin{bmatrix}1\\0\end{bmatrix}$$

Use this information to find the solution to $A\mathbf{x}=\begin{bmatrix}0\\3\end{bmatrix}$.

A. $\mathbf{x}=\begin{bmatrix}3\\-7\end{bmatrix}$

B. $\mathbf{x}=\begin{bmatrix}-3\\7\end{bmatrix}$

C. $\mathbf{x}=\begin{bmatrix}4\\-1\end{bmatrix}$

D. $\mathbf{x}=\begin{bmatrix}2\\-6\end{bmatrix}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

## Q #6 Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following statements is/are **TRUE**?

A. If a linear system has strictly more equations than variables, it must be inconsistent.

B. If $A$ is an $m\times n$ matrix such that $A\mathbf{x}=0$ has only the trivial solution, then $A\mathbf{x}=\mathbf{b}$ has at most one solution for every $\mathbf{b}\in\mathbb{R}^m$.

C. The rows of any $4\times 5$ matrix are linearly dependent.

D. If the linear system $A\mathbf{x}=\mathbf{b}$ is inconsistent, then the coefficient matrix $A$ does not have a pivot position in every row.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

## Q #7 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

For which vector $\mathbf{b}$ below does the system $\begin{bmatrix}2&4\\4&6\\3&4\end{bmatrix}\mathbf{x}=\mathbf{b}$ have a solution?

A. $\begin{bmatrix}2\\3\\1\end{bmatrix}$

B. $\begin{bmatrix}3\\4\\3\end{bmatrix}$

C. $\begin{bmatrix}1\\2\\1\end{bmatrix}$

D. $\begin{bmatrix}2\\2\\1\end{bmatrix}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

## Q #8 Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider the following statements about linear systems and matrix theory. Which of the following statements are TRUE?

A. If every column of the coefficient matrix of a linear system $A\mathbf{x} = \mathbf{b}$ contains a pivot, then the system has at most one solution.
B. If the coefficient matrix of a linear system $A\mathbf{x} = \mathbf{b}$ has a pivot in every row, then the system has at least one solution.
C. If the linear system $A\mathbf{x} = \mathbf{0}$ has at least one solution, then $A\mathbf{x} = \mathbf{b}$ must have at least one solution.
D. If the linear system $A\mathbf{x} = \mathbf{0}$ has at most one solution, then $A\mathbf{x} = \mathbf{b}$ has at most one solution.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider the following statements about linear systems and matrices. Which of the following is/are TRUE?

A. If $R$ is the reduced row echelon form of an $m \times n$ matrix $A$ and $A\mathbf{x} = \mathbf{b}$ is consistent for some vector $\mathbf{b} \in \mathbb{R}^m$, then $R\mathbf{x} = \mathbf{b}$ must be consistent.
B. If the linear system $A\mathbf{x} = \mathbf{b}$ has more than one solution for some value $\mathbf{b}$, then the linear system $A\mathbf{x} = 0$ also has more than one solution.
C. If every column of a matrix has a pivot, then the system is always consistent.
D. If every row of a matrix has a pivot, then the system is always consistent.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let $Ax = b$ be a system of linear equations where $A$ is an $m \times n$ matrix and $b$ is a $m \times 1$ column vector and $X$ is an $n \times 1$ column vector of unknowns. Which of the following is/are $\mathbf{True}$?

A. The system has a solution if and only if, both $A$ and the augmented matrix $[Ab]$ have the same rank.

B. If $m < n$ and $b$ is the zero vector, then the system has infinitely many solutions.

C. If $m = n$ and $b$ is a non-zero vector, then the system has a unique solution.

D. The system will have only a trivial solution when $m = n$, $b$ is the zero vector and $\text{rank}(A) = n$.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

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